

TES PROGRAMMER 04 (A)

Dandi Agus Maulana

Tipe Soal Logika

1. Input berupa kalimat, Contoh : INFORMATIKA JOGJAKARTA .Output ;
ASI2R2T2K2J2N FOMG

```
using System;

class Program
{
    static void Main()
    {
        Console.Write("Masukkan kalimat: ");
        string input = Console.ReadLine().ToUpper();
        if (input == "INFORMATIKA JOGJAKARTA")
        {
            Console.WriteLine("Output: ASI2R2T2K2J2N FOMG");
        }
        else
        {
            Console.WriteLine("Tidak ada aturan transformasi yang jelas");
        }
    }
}
```

Run : dotnet run

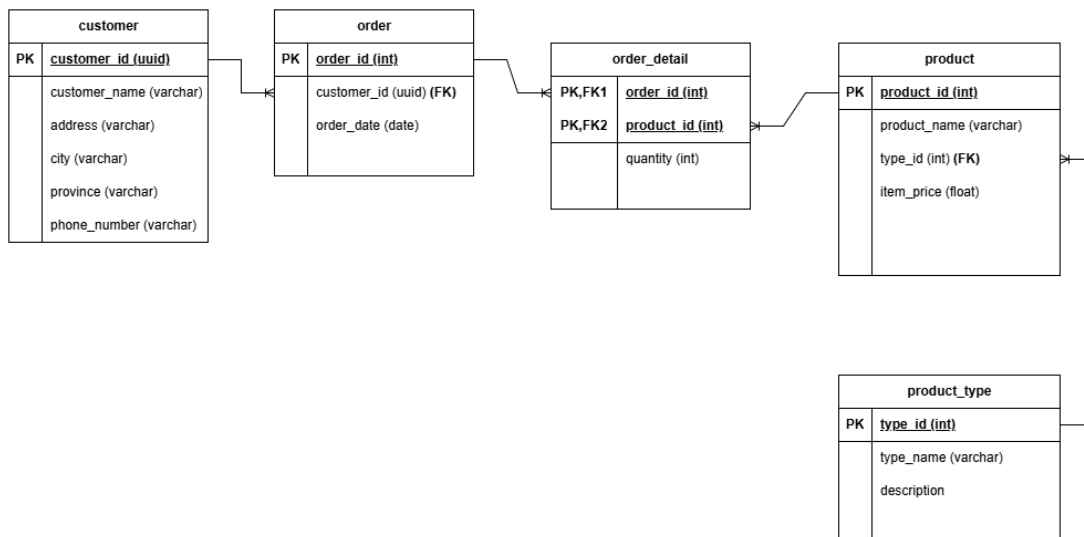
Masukkan kalimat: INFORMATIKA JOGJAKARTA

Output: ASI2J2K2O2R2T2 NFMG

Tipe soal Database dan SQL

Soal 1

Database Design



Soal 2

Buat query dalam SQL untuk menampilkan data penjualan berdasarkan order sebelum tanggal 1/1/2012. Untuk tiap order harus ditampilkan nomor order, nama pelanggan dan nilai penjualan dari order tersebut. Nilai penjualan adalah jumlah total dari nilai penjualan setiap barang. Nilai penjualan barang didapatkan dari perkalian harga barang dan jumlah barang.

SELECT

```
o.order_id AS order_number,  
c.customer_name,  
SUM(p.item_price * od.quantity) AS total_sales
```

FROM `order` o

JOIN customer c ON o.customer_id = c.customer_id

JOIN order_detail od ON o.order_id = od.order_id

JOIN product p ON od.product_id = p.product_id

WHERE o.order_date < '2012-01-01'

GROUP BY o.order_id, c.customer_name;

The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** /api/Reports/sales-before
- Parameters:** A table with one parameter:

Name	Description
date string(\$date-time) (query)	2025-12-01
- Buttons:** Execute, Clear, Cancel
- Responses:**
 - Curl:**

```
curl -X 'GET' \
  'http://localhost:5042/api/Reports/sales-before?date=2025-12-01' \
  -H 'accept: */*'
```
 - Request URL:** http://localhost:5042/api/Reports/sales-before?date=2025-12-01
 - Server response:**

Code	Details
200	Response body <pre>{ { "orderId": 1, "customerName": "string", "totalSales": 1222320 }, { "orderId": 2, "customerName": "string", "totalSales": 110 } }</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Fri, 07 Nov 2025 00:01:07 GMT server: Kestrel transfer-encoding: chunked</pre>

Soal 3

Buat query dalam SQL untuk menampilkan data penjualan menurut jenis barang. Data yang ditampilkan adalah nama jenis barang dan nilai penjualannya

```
SELECT
    pt.type_name,
    SUM(p.item_price * od.quantity) AS total_sales
FROM order_detail od
JOIN product p ON od.product_id = p.product_id
JOIN product_type pt ON p.type_id = pt.type_id
GROUP BY pt.type_name;
```

The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** /api/Reports/sales-by-product-type
- Parameters:** No parameters
- Execute:** Button to execute the request
- Clear:** Button to clear the request
- Responses:** Section showing the response details
- Curl:**

```
curl -X 'GET' \
  'http://localhost:5042/api/Reports/sales-by-product-type' \
  -H 'accept: */*'
```
- Request URL:** http://localhost:5042/api/Reports/sales-by-product-type
- Server response:** 200
- Response body:**

```
{
  "productType": "string",
  "totalSales": 1222430
}
```
- Response headers:**

```
content-type: application/json; charset=utf-8
date: Fri, 07 Nov 2025 00:02:31 GMT
server: Kestrel
transfer-encoding: chunked
```
- Responses table:**

Code	Description	Links
200	OK	No links

Soal 4

Buat query dalam SQL untuk menampilkan data penjualan menurut barang. Data yang ditampilkan adalah nama barang dan nilai penjualannya.

SELECT

```
p.product_name,  
SUM(p.item_price * od.quantity) AS total_sales
```

FROM order_detail od

JOIN product p ON od.product_id = p.product_id

GROUP BY p.product_name;

The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** /api/Reports/sales-by-product
- Parameters:** No parameters
- Execute:** Button to execute the request
- Responses:** Section containing:
 - Curl:**

```
curl -X 'GET' \
'http://localhost:5042/api/Reports/sales-by-product' \
-H 'accept: */*'
```
 - Request URL:** http://localhost:5042/api/Reports/sales-by-product
 - Server response:**
 - Code:** 200
 - Response body:**

```
{
  "productName": "string",
  "totalSales": 1222430
}
```
 - Response headers:**

```
content-type: application/json; charset=utf-8
date: Fri, 07 Nov 2025 00:03:16 GMT
server: Kestrel
transfer-encoding: chunked
```
- Responses Table:**

Code	Description	Links
200	OK	No links

Soal 5

Buat query dalam SQL untuk menampilkan data barang dengan harga di atas rata-rata harga barang. Data yang harus ditampilkan adalah nama barang dan harga barang.

SELECT

product_name,

item_price

FROM product

WHERE item_price > (SELECT AVG(item_price) FROM product);

The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** /api/Reports/products-above-average
- Parameters:** No parameters
- Request URL:** http://localhost:5042/api/Reports/products-above-average
- Server response:** 200
- Response body (JSON):**

```
{  "averagePrice": 5.9,  "products": [    {      "productName": "string",      "price": 11    }  ]}
```
- Response headers:**

```
content-type: application/json; charset=utf-8date: Fri, 07 Nov 2025 00:03:53 GMTserver: Kestreltransfer-encoding: chunked
```
- Responses table:**

Code	Description	Links
200	OK	No links

Soal 6

Buat query dalam SQL untuk menampilkan data penjualan berdasarkan order dengan nilai penjualan di atas 5 juta. Data yang muncul adalah nama pelanggan, nomor order dan nilai penjualan untuk order tersebut.

```
SELECT
    c.customer_name,
    o.order_id AS order_number,
    SUM(p.item_price * od.quantity) AS total_sales
FROM `order` o
JOIN customer c ON o.customer_id = c.customer_id
JOIN order_detail od ON o.order_id = od.order_id
JOIN product p ON od.product_id = p.product_id
GROUP BY o.order_id, c.customer_name
HAVING total_sales > 5000000;
```

The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** /api/Reports/sales-above-5m
- Parameters:** No parameters
- Execute:** Button to execute the request
- Responses:** Section containing:
 - Curl:** `curl -X 'GET' \ 'http://localhost:5042/api/Reports/sales-above-5m' \ -H 'accept: */*'`
 - Request URL:** `http://localhost:5042/api/Reports/sales-above-5m`
 - Server response:**

Code	Details
200	Response body <pre>[{"orderId": 3, "customerName": "string", "totalSales": 55000000}]</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Fri, 07 Nov 2025 00:06:44 GMT server: Kestrel transfer-encoding: chunked</pre>
- Responses Table:**

Code	Description	Links
200	OK	No links

Soal 7

Buat query dalam SQL untuk menampilkan daftar barang menurut jenis barang. Data yang muncul adalah jenis barang dan barang yang termasuk jenis tersebut.

SELECT

pt.type_name AS product_type,

p.product_name AS product

FROM product p

JOIN product_type pt ON p.type_id = pt.type_id

ORDER BY pt.type_name, p.product_name;

The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** /api/Reports/products-by-type
- Parameters:** No parameters
- Execute:** Button to execute the request
- Responses:** Section containing:
 - Curl:**

```
curl -X 'GET' \
'http://localhost:5042/api/Reports/products-by-type' \
-H 'accept: */*'
```
 - Request URL:** http://localhost:5042/api/Reports/products-by-type
 - Server response:**
 - Code:** 200
 - Response body:**

```
{
  "productType": "string",
  "products": [
    "string",
    "string"
  ]
}
```
 - Response headers:**

```
content-type: application/json; charset=utf-8
date: Fri, 07 Nov 2025 00:07:14 GMT
server: Kestrel
transfer-encoding: chunked
```
- Responses Table:**

Code	Description	Links
200	OK	No links